AP CALCULUS AB · EXAMINATION REPOSITORY

Unit 8
Applications of Integration

100-Question Master Practice Set

- Questions 1–90: Multiple Choice
- Questions 91–100: Free Response
- Average value, motion, area between curves, cross sections, disks & washers
- Priority stars, diagnostic answer key, detailed step-by-step solutions, and AP rubrics

Original AP-aligned practice · Moderate focus · Not an official College Board paper

Exam Structure & Priority Guide

The question paper is continuously numbered Q1–Q100. The answer section begins only after all questions.

Priority Ratings

★★★★★	**Very Important**	Average value, accumulation, net change, area, cross sections, and disk/washer setups.
★★★	**Mandatory**	Cross sections, rectilinear motion, FTC, and standard revolution calculations.
★★	**Important**	Foundational definite-integral and geometric-area practice.

- **Q1–Q90:** Four-option MCQs with one intended correct answer.
- **Q91–Q100:** Multi-part FRQs with graphs/tables where appropriate; 9 points each.
- **Unit 8 topics:** 8.1 Average Value; 8.2 Position, Velocity & Acceleration; 8.3 Linear Accumulation; 8.4 Area Between Curves (dx); 8.5 Area Between Curves (dy); 8.6 Known Cross Sections; 8.7 Disk Method; 8.8 Washer Method.
- **Scope:** Original AP-style practice—not reproduced College Board exam questions.

AP Calculus AB Unit 8 Topic Coverage

8.1 Average Value of a Function

8.2 Position, Velocity, and Acceleration

8.3 Accumulation Functions

8.4 Area Between Curves (Vertical Slices)

8.5 Area Between Curves (Horizontal Slices)

8.6 Volumes: Known Cross Sections

8.7 Volumes: Disk Method

8.8 Volumes: Washer Method

Directions: Attempt the full paper before reading solutions. For volume questions, identify the cross-sectional shape or the outer and inner radii before integrating.

QUESTION 01

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The average value of $f(x)=x^2$ on $[0,3]$ is

A 9
B 3
C 6
D 2

QUESTION 02

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The average value of $f(x)=2x+1$ on $[1,4]$ is

A 4
B 5
C 7
D 6

QUESTION 03

EASY · MULTIPLE CHOICE ★★ IMPORTANT

The average value of $f(x)=4-x$ on $[0,4]$ is

A 3
B 2
C 1
D 4

QUESTION 04

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The average value of sin x on $[0,\pi]$ is

A $1/\pi$
B 2
C $2/\pi$
D 1

QUESTION 05

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The average value of $1/x$ on $[1,e]$ is

A $1/(e-1)$
B $e-1$
C $1/e$
D 1

QUESTION 06

EASY · MULTIPLE CHOICE ★★ IMPORTANT

The average value of the constant function $f(x)=7$ on any nonzero-length interval is

A 0
B 1
C 7
D the interval length

QUESTION 07

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The average value of x^2 on $[-1,2]$ is

A 0

B 3

C 1

D 2

QUESTION 08

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The average value of e^x on $[0,\ln 3]$ is

A 1

B $2/(\ln 3)$

C $2\ln 3$

D $1/(\ln 3)$

QUESTION 09

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The average value of x^3 on $[-2,2]$ is

A 0

B 2

C 8

D 4

QUESTION 10

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $f(x)=x^2$ on $[0,4]$, the value c satisfying $f(c)$=the average value of f is

A $16/3$

B $1/\sqrt{3}$

C $4/\sqrt{3}$

D $2\sqrt{3}$

QUESTION 11

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A particle has $v(t)=t-2$ on $[0,4]$. Its displacement is

A 4

B 0

C 2

D 8

QUESTION 12

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $v(t)=t-2$ on $[0,4]$, the total distance traveled is

A 8

B 4

C 2

D 6

QUESTION 13

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

A particle has $v(t)=t^2-4$ on $[0,3]$. Its displacement is

A. 3
B. 0
C. -3
D. $23/3$

QUESTION 14

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $v(t)=t^2-4$ on $[0,3]$, the total distance traveled is

A. -3
B. $16/3$
C. $7/3$
D. $23/3$

QUESTION 15

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

A particle has $s(t)=t^3-3t^2$. It is at rest at

A. $t=0$ only
B. $t=2$ only
C. $t=0$ and $t=2$
D. $t=1$ and $t=3$

QUESTION 16

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

A particle has $a(t)=6t-4$ and $v(0)=3$. Then $v(2)$ is

A 3

B 7

C 11

D 5

QUESTION 17

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Velocity values at $t=0,1,2,3$ are $2,0,-2,0$. If v is linear between listed points, displacement on $[0,3]$ is

A 3

B 0

C -1

D 1

QUESTION 18

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $s(t)=t^2-4t$, the average velocity on $[0,3]$ is

A 1

B -1

C 3

D -3

QUESTION 19

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A particle has $v(t)=3t^2-12t+9$. It moves in the negative direction when

A) all $t>0$
B) 1
C) 0
D) $t>3$

QUESTION 20

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $v<0$ and $a>0$ at an instant, the particle is

A) at rest
B) moving left and speeding up
C) moving left and slowing down
D) moving right and speeding up

QUESTION 21

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $F(x)=\int_0^x(t^2+1)dt$, then $F'(x)=$

A) $x^3/3+x$
B) $2x+1$
C) x^2+1
D) $2x$

QUESTION 22

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $G(x)=\int_0^{(x^2)} \cos(t)dt$, then $G'(x)=$

A $\cos(x^2)$

B $2x\sin(x^2)$

C $-2x\sin(x^2)$

D $2x\cos(x^2)$

QUESTION 23

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $H(x)=\int_x^4 (t^2+1)dt$, then $H'(x)=$

A x^2+1

B $-(x^2+1)$

C 4^2+1

D $2x$

QUESTION 24

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $J(x)=\int_x^{(x^2)} \sin(t)dt$, then $J'(x)=$

A $2x\sin(x^2)-\sin(x)$

B $\sin(x^2)-\sin(x)$

C $2x\sin(x^2)+\sin(x)$

D $2x\cos(x^2)-\cos(x)$

QUESTION 25

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If $F(x)=\int_0^x(t-1)dt$, then $F(x)=$

A $x^2/2-1$

B $x^2/2-x$

C x^2-1

D $x-1$

QUESTION 26

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $F(x)=\int_0^x(t^2-4)dt$, the critical numbers of F are

A $x=0$ only

B none

C $x=-4$ and $x=4$

D $x=-2$ and $x=2$

QUESTION 27

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $F(x)=\int_0^x(t^2-4)dt$, F is increasing when

A -2

B $x>0$

C all real x

D $x<-2$ or $x>2$

QUESTION 28

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int_1^3 2x\,dx =$

A 10

B 4

C 6

D 8

QUESTION 29

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\int_0^(\pi/2) \cos x\,dx =$

A $\pi/2$

B 0

C 1

D 2

QUESTION 30

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\int_0^2 t^4\,dt =$

A 8

B 16/5

C 16

D 32/5

QUESTION 31

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The area between $y=x$ and $y=x^2$ on $[0,1]$ is

A 1/2
B 1/3
C 2/3
D 1/6

QUESTION 32

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $y=2x$ and $y=x^2$ on $[0,2]$ is

A 8/3
B 2/3
C 4
D 4/3

QUESTION 33

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $y=4$ and $y=x^2$ on $[-2,2]$ is

A 32/3
B 8
C 16
D 16/3

QUESTION 34

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The area enclosed by $y=x+2$ and $y=x^2$ from $x=-1$ to $x=2$ is

A 6
B 9
C 3
D 9/2

QUESTION 35

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $y=\sqrt{x}$ and $y=x/2$ on $[0,4]$ is

A 4/3
B 8/3
C 2
D 2/3

QUESTION 36

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $y=x$ and $y=x^3$ on $[0,1]$ is

A 1/2
B 1/12
C 1/4
D 1/3

QUESTION 37

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $y=1$ and $y=x^2$ on $[-1,1]$ is

A 1/3
B 4/3
C 2/3
D 2

QUESTION 38

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The area enclosed by $y=2-x$ and $y=x^2$ from $x=-2$ to $x=1$ is

A 9/2
B 15/2
C 6
D 3

QUESTION 39

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $y=3x$ and $y=x^2$ on $[0,3]$ is

A 9
B 9/2
C 6
D 3

QUESTION 40

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $y=1$ and $y=|x|$ on $[-1,1]$ is

A 1
B 3/2
C 1/2
D 2

QUESTION 41

EASY · MULTIPLE CHOICE ★★ IMPORTANT

The area between $y=\cos x$ and the x-axis on $[0,\pi/2]$ is

A 0
B 1
C $\pi/2$
D 2

QUESTION 42

EASY · MULTIPLE CHOICE ★★ IMPORTANT

The area between $y=2$ and $y=x$ on $[0,2]$ is

A 3
B 2
C 4
D 1

QUESTION 43

EASY · MULTIPLE CHOICE

★★ IMPORTANT

The area between $y=4$ and $y=x$ on $[0,4]$ is

A 16

B 6

C 8

D 4

QUESTION 44

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The area between $y=6-x$ and $y=x$ on $[0,3]$ is

A 9

B 6

C 12

D 18

QUESTION 45

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The area between $y=3$ and $y=x^2$ on $[0,\sqrt{3}]$ is

A $\sqrt{3}$

B $2\sqrt{3}$

C $3\sqrt{3}$

D $4\sqrt{3}$

QUESTION 46

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The area between $x=2y$ and $x=y^2$ for $0\le y\le 2$ is

A 2/3

B 4/3

C 8/3

D 2

QUESTION 47

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $x=y$ and $x=y^2$ for $0\le y\le 1$ is

A 1/2

B 1/6

C 2/3

D 1/3

QUESTION 48

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area bounded by $x=0$ and $x=4-y^2$ for $-2\le y\le 2$ is

A 16/3

B 32/3

C 8

D 16

QUESTION 49

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $x=y^2$ and $x=3$ for $0\le y\le\sqrt{3}$ is

A $4\sqrt{3}$
B $3\sqrt{3}$
C $\sqrt{3}$
D $2\sqrt{3}$

QUESTION 50

EASY · MULTIPLE CHOICE ★★ IMPORTANT

The area between $x=y$ and $x=2$ for $0\le y\le 2$ is

A 2
B 3
C 4
D 1

QUESTION 51

EASY · MULTIPLE CHOICE ★★ IMPORTANT

The area bounded by $x=0$ and $x=4-y$ for $0\le y\le 4$ is

A 8
B 4
C 6
D 16

QUESTION 52

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $x=0$ and $x=\sqrt{y}$ for $0\le y\le 4$ is

A $16/3$

B 16

C 8

D 4

QUESTION 53

EASY · MULTIPLE CHOICE ★★ IMPORTANT

The area between $x=0$ and $x=1+y$ for $0\le y\le 2$ is

A 3

B 4

C 6

D 2

QUESTION 54

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The area between $x=0$ and $x=3$ for $-\sqrt{3}\le y\le\sqrt{3}$ is

A $2\sqrt{3}$

B $4\sqrt{3}$

C $8\sqrt{3}$

D $6\sqrt{3}$

QUESTION 55

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The area between $x=y+2$ and $x=y^2$ for $-1\le y\le 2$ is

A 9/2

B 6

C 15/2

D 3

QUESTION 56

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A solid has square cross sections perpendicular to the x-axis. Each square's side is $\sqrt{x}$ for $0\le x\le 4$. Its volume is

A 16

B 8

C 4

D 6

QUESTION 57

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Square cross sections perpendicular to the x-axis have side $2-x$ on $0\le x\le 2$. The volume is

A 4/3

B 8

C 4

D 8/3

QUESTION 58

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Square cross sections have side $1-x^2$ for $-1 \le x \le 1$. The volume is

A $8/15$

B 2

C $4/3$

D $16/15$

QUESTION 59

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Rectangular cross sections have base x and height twice the base, for $0 \le x \le 1$. The volume is

A 2

B $1/3$

C $2/3$

D 1

QUESTION 60

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Equilateral-triangle cross sections have side $1-x$ for $0 \le x \le 1$. The volume is

A $\sqrt{3}/3$

B $\sqrt{3}/24$

C $\sqrt{3}/12$

D $\sqrt{3}/6$

QUESTION 61

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Semicircular cross sections have diameter $2-x$ for $0\le x\le 2$. The volume is

A $\pi/6$
B $2\pi/3$
C $\pi/3$
D π

QUESTION 62

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Semicircular cross sections have diameter $\sqrt{(1-x^2)}$ for $-1\le x\le 1$. The volume is

A $\pi/6$
B $\pi/3$
C $\pi/12$
D $2\pi/3$

QUESTION 63

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Right-isosceles-triangle cross sections have a leg of length $2-x$ for $0\le x\le 2$. The volume is

A $2/3$
B $8/3$
C $4/3$
D 4

QUESTION 64

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Equilateral-triangle cross sections have side 2x for 0≤x≤1. The volume is

A $\sqrt{3}$

B $2\sqrt{3}/3$

C $\sqrt{3}/3$

D $\sqrt{3}/6$

QUESTION 65

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Rectangular cross sections have base 1−x and height 3 times the base for 0≤x≤1. The volume is

A 1

B 1/2

C 3

D 3/2

QUESTION 66

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Square cross sections have side 2x for 0≤x≤1. The volume is

A 4

B 4/3

C 2/3

D 1/3

QUESTION 67

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Semicircular cross sections have diameter $2\sqrt{x}$ for $0 \le x \le 1$. The volume is

A $\pi/8$

B $\pi/2$

C $\pi/4$

D π

QUESTION 68

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Square cross sections have side $\sqrt{(4-x^2)}$ for $-2 \le x \le 2$. The volume is

A 16

B 8/3

C 16/3

D 32/3

QUESTION 69

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region under $y=x$ on $0 \le x \le 2$ is revolved about the x-axis. The volume is

A $8\pi/3$

B π

C 8π

D 4π

QUESTION 70

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region under $y=\sqrt{x}$ on $0\le x\le 4$ is revolved about the x-axis. The volume is

A $16\pi/3$

B 8π

C 16π

D 4π

QUESTION 71

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The region under $y=3$ on $0\le x\le 2$ is revolved about the x-axis. The volume is

A 9π

B 6π

C 18π

D 36π

QUESTION 72

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region under $y=1-x^2$ on $-1\le x\le 1$ is revolved about the x-axis. The volume is

A $8\pi/15$

B $4\pi/3$

C 2π

D $16\pi/15$

QUESTION 73

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region between $x=0$ and $x=\sqrt{y}$ for $0\le y\le 4$ is revolved about the y-axis. The volume is

A $16\pi/3$

B 4π

C $8\pi/3$

D 8π

QUESTION 74

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region between $x=0$ and $x=y^2$ for $0\le y\le 2$ is revolved about the y-axis. The volume is

A $32\pi/5$

B 8π

C $16\pi/5$

D 32π

QUESTION 75

EASY · MULTIPLE CHOICE ★★ IMPORTANT

The region under $y=x$ on $0\le x\le 1$ is revolved about the x-axis. The volume is

A $\pi/6$

B π

C $\pi/2$

D $\pi/3$

QUESTION 76

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The region under $y=2x$ on $0\le x\le 1$ is revolved about the x-axis. The volume is

A $2\pi/3$

B $\pi/3$

C $4\pi/3$

D 4π

QUESTION 77

EASY · MULTIPLE CHOICE ★★ IMPORTANT

The region under $y=4$ on $0\le x\le 3$ is revolved about the x-axis. The volume is

A 24π

B 48π

C 12π

D 64π

QUESTION 78

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region between $x=0$ and $x=1+y/2$ for $0\le y\le 2$ is revolved about the y-axis. The volume is

A 6π

B $16\pi/3$

C $14\pi/3$

D $10\pi/3$

QUESTION 79

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The region under $y=2-x^2$ on $0\le x\le\sqrt{2}$ is revolved about the x-axis. The volume is

A $32\sqrt{2}\pi/15$

B $4\sqrt{2}\pi$

C $8\sqrt{2}\pi$

D $16\sqrt{2}\pi/15$

QUESTION 80

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The region between $y=2$ and $y=x$ on $0\le x\le 2$ is revolved about the x-axis. The volume is

A 8π

B $8\pi/3$

C 16π

D $16\pi/3$

QUESTION 81

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The region between $y=\sqrt{x}$ and $y=x$ on $0\le x\le 1$ is revolved about the x-axis. The volume is

A $\pi/2$

B $\pi/3$

C $\pi/6$

D $\pi/12$

QUESTION 82

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The region between $y=3$ and $y=1$ on $0\le x\le 2$ is revolved about the x-axis. The volume is

A 18π

B 20π

C 8π

D 16π

QUESTION 83

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region between $y=2x$ and $y=x^2$ on $0\le x\le 2$ is revolved about the x-axis. The volume is

A $32\pi/15$

B $8\pi/3$

C $16\pi/3$

D $64\pi/15$

QUESTION 84

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The region between $y=4$ and $y=\sqrt{x}$ on $0\le x\le 4$ is revolved about the x-axis. The volume is

A 72π

B 56π

C 48π

D 64π

QUESTION 85

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region between $y=0$ and $y=x$ on $0\le x\le 1$ is revolved about $y=-1$. The volume is

A 4π
B π
C $4\pi/3$
D $2\pi/3$

QUESTION 86

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region under $y=x$ on $0\le x\le 2$ is revolved about $y=3$. The volume is

A $28\pi/3$
B 18π
C $32\pi/3$
D $20\pi/3$

QUESTION 87

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region between $x=y$ and $x=2$ on $0\le y\le 2$ is revolved about the y-axis. The volume is

A 8π
B $16\pi/3$
C 16π
D $8\pi/3$

QUESTION 88

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region between $x=1+y$ and $x=2$ on $0\le y\le 1$ is revolved about the y-axis. The volume is

A $5\pi/3$

B $16\pi/3$

C $10\pi/3$

D $8\pi/3$

QUESTION 89

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The region between $y=1$ and $y=x^2$ on $-1\le x\le 1$ is revolved about the x-axis. The volume is

A 2π

B $4\pi/5$

C π

D $8\pi/5$

QUESTION 90

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The region between $y=2$ and $y=1$ on $0\le x\le 3$ is revolved about the x-axis. The volume is

A 6π

B 12π

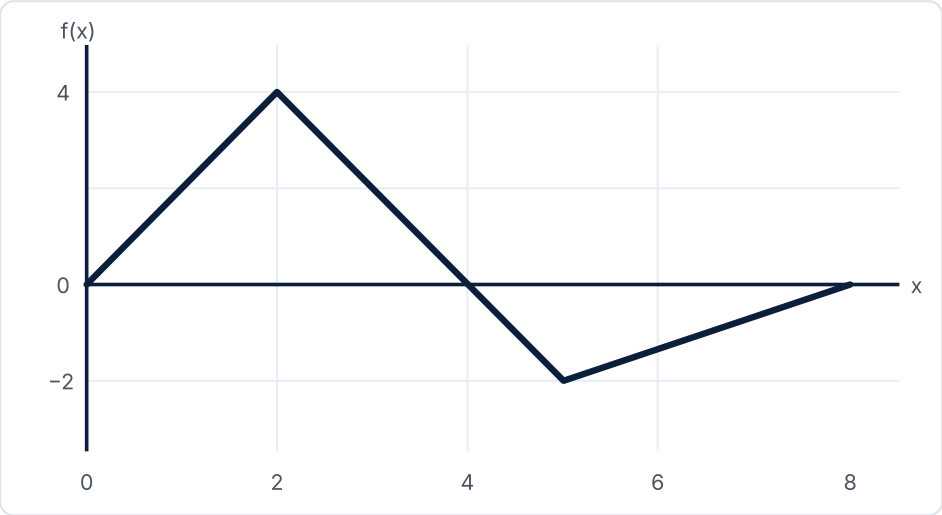
C 9π

D 3π

Question 91: Area and Net Area from a Graph

★★★★★ Very Important | 9 points

The graph of f is made of straight segments through the plotted vertices on $0 \le x \le 8$.



The graph is formed by straight segments through the plotted vertices (0,0), (2,4), (5,−2), and (8,0).

(a) [3 points] Evaluate $\int_0^8 f(x)\,dx$ using signed areas.

(b) [2 points] Find the total area between the graph and the x-axis.

(c) [2 points] Find the average value of f on [0,8].

(d) [2 points] Explain why the values in (a) and (b) differ.

Show your setup. Include limits of integration, cross-sectional area or washer radii, and units when relevant.

Question 92: Average Value and the Integral Mean Value Theorem

★★★★★ Very Important 9 points

Let $f(x)=x^2$ on $[0,3]$.

(a) [3 points] Find the average value of f on [0,3].

(b) [3 points] Find c in [0,3] such that f(c) equals the average value.

(c) [3 points] State why the Integral Mean Value Theorem guarantees that at least one such c exists.

Show your setup. Include limits of integration, cross-sectional area or washer radii, and units when relevant.

Question 93: Motion: Displacement and Total Distance

★★★★★ Very Important | 9 points

A particle moves along a line with velocity $v(t)=t^2-4t+3$ meters per second for $0\le t\le 5$.

(a) [3 points] Find the displacement of the particle on $[0,5]$.

(b) [4 points] Find the total distance traveled on $[0,5]$.

(c) [2 points] State the intervals on which the particle moves in the positive direction.

Show your setup. Include limits of integration, cross-sectional area or washer radii, and units when relevant.

Question 94: Area Between Curves

★★★★★ Very Important | 9 points

Consider the region bounded by $y=2x$ and $y=x^2$ for $0 \le x \le 2$.

(a) [2 points] Which curve is above the other on the interior of the interval? Justify briefly.

(b) [4 points] Set up and evaluate the area of the region.

(c) [3 points] Find the average vertical distance between the curves on $[0,2]$.

Show your setup. Include limits of integration, cross-sectional area or washer radii, and units when relevant.

Question 95: Volume with Square Cross Sections

★★★★★ Very Important | 9 points

The base of a solid is the region under $y=\sqrt{x}$ and above the x-axis for $0\le x\le 4$. Cross sections perpendicular to the x-axis are squares.

(a) [2 points] Write the side length of a cross section at x.

(b) [2 points] Write the cross-sectional area A(x).

(c) [5 points] Set up and calculate the volume.

Show your setup. Include limits of integration, cross-sectional area or washer radii, and units when relevant.

Question 96: Volume with Semicircular Cross Sections

★★★★★ Very Important | 9 points

The base is the region under $y=\sqrt{(1-x^2)}$ and above the x-axis for $-1\le x\le 1$. Cross sections perpendicular to the x-axis are semicircles whose diameters are the vertical segments in the base.

(a) [2 points] State the diameter and radius of a cross section at x.

(b) [3 points] Find the area A(x) of a semicircular cross section.

(c) [4 points] Set up and evaluate the volume.

Show your setup. Include limits of integration, cross-sectional area or washer radii, and units when relevant.

Question 97: Disk Method About the x-Axis

★★★★★ Very Important | 9 points

The region bounded by $y=\sqrt{x}$, the x-axis, $x=0$, and $x=4$ is revolved about the x-axis.

(a) [2 points] State the radius of a disk at x.

(b) [3 points] Set up the disk-method integral for the volume.

(c) [4 points] Evaluate the volume exactly.

Show your setup. Include limits of integration, cross-sectional area or washer radii, and units when relevant.

Question 98: Washer Method About the x-Axis

★★★★★ Very Important | 9 points

The region between $y=2$ and $y=x$ on $0\le x\le 2$ is revolved about the x-axis.

(a) [2 points] Identify the outer and inner radii of a washer.

(b) [3 points] Set up the washer-method integral.

(c) [4 points] Evaluate the volume exactly.

Show your setup. Include limits of integration, cross-sectional area or washer radii, and units when relevant.

Question 99: Volume with Cross Sections from a Table

★★★ Mandatory 9 points

A base region has cross sections perpendicular to the x-axis. The table gives the base-segment length s(x). Each cross section is a square.

x	0	1	2	3
s(x)	1	2	1	0

(a) [3 points] Use a left Riemann sum with n=3 to estimate the volume.

(b) [3 points] Use a right Riemann sum with n=3 to estimate the volume.

(c) [3 points] Explain why the cross-sectional area is $s(x)^2$, not s(x).

Show your setup. Include limits of integration, cross-sectional area or washer radii, and units when relevant.

Question 100: Washer Method About a Shifted Axis

★★★★★ Very Important | 9 points

The region under $y=x$ on $0\le x\le 2$ is revolved about the horizontal line $y=3$.

(a) [2 points] At a given x, identify the outer and inner radii.

(b) [3 points] Write the washer-method integral.

(c) [4 points] Evaluate the exact volume.

Show your setup. Include limits of integration, cross-sectional area or washer radii, and units when relevant.

Questions 1–100 Complete

The next pages contain the MCQ answer key, explanations for every MCQ distractor, and FRQ model solutions with scoring rubrics.

> **Practice scoring:** 90 MCQs at 1 point each plus 10 FRQs at 9 points each gives 180 raw points. This is only a practice scheme, not an official AP score conversion.

Multiple-Choice Answer Key

Letters reflect the deterministic shuffled choices shown in the question paper.

Q01 B ★★★	Q02 D ★★★	Q03 B ★★	Q04 C ★★★	Q05 A ★★★	Q06 C ★★	Q07 C ★★★	Q08 B ★★★	Q09 A ★★★	Q10 C ★★★★★
Q11 B ★★★★★	Q12 B ★★★★★	Q13 C ★★★	Q14 D ★★★★★	Q15 C ★★★	Q16 B ★★★	Q17 C ★★★	Q18 B ★★★	Q19 B ★★★★★	Q20 C ★★★★★
Q21 C ★★★★★	Q22 D ★★★★★	Q23 B ★★★★★	Q24 A ★★★★★	Q25 B ★★★	Q26 D ★★★★★	Q27 D ★★★★★	Q28 D ★★	Q29 C ★★	Q30 D ★★★
Q31 D ★★★★★	Q32 D ★★★	Q33 A ★★★	Q34 D ★★★★★	Q35 A ★★★	Q36 C ★★★	Q37 B ★★★	Q38 A ★★★★★	Q39 B ★★★	Q40 A ★★★
Q41 B ★★	Q42 B ★★	Q43 C ★★	Q44 A ★★★	Q45 B ★★★	Q46 B ★★★★★	Q47 B ★★★	Q48 B ★★★	Q49 D ★★★	Q50 A ★★
Q51 A ★★	Q52 A ★★★	Q53 B ★★	Q54 D ★★★	Q55 A ★★★★★	Q56 B ★★★★★	Q57 D ★★★	Q58 D ★★★★★	Q59 C ★★★	Q60 C ★★★
Q61 C ★★★★★	Q62 A ★★★★★	Q63 C ★★★	Q64 C ★★★	Q65 A ★★★	Q66 B ★★★	Q67 C ★★★	Q68 D ★★★★★	Q69 A ★★★★★	Q70 B ★★★★★
Q71 C ★★★	Q72 D ★★★★★	Q73 D ★★★★★	Q74 A ★★★★★	Q75 D ★★	Q76 C ★★★	Q77 B ★★	Q78 C ★★★★★	Q79 A ★★★★★	Q80 D ★★★★★
Q81 C ★★★★★	Q82 D ★★★	Q83 D ★★★★★	Q84 B ★★★	Q85 C ★★★★★	Q86 A ★★★★★	Q87 B ★★★★★	Q88 A ★★★★★	Q89 D ★★★★★	Q90 C ★★★

Question 01

★★★ Mandatory

Correct answer: (B) 3

STEP-BY-STEP SOLUTION

Average value = $(1/3)\int_0^3 x^2 dx = (1/3)(9) = 3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses f(3) as the average.

✗ **(C)** Forgets to divide by interval length 3.

✗ **(D)** Uses an incorrect antiderivative.

Question 02

★★★ Mandatory

Correct answer: (D) 6

STEP-BY-STEP SOLUTION

$(1/3)\int_1^4 (2x+1)dx = (1/3)[x^2+x]_1^4 = (1/3)(20-2) = 6$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Underestimates the integral average.

✗ **(B)** Uses an incorrect interval length.

✗ **(C)** Uses the value at x=3 only.

Question 03

★★ Important

Correct answer: (B) 2

STEP-BY-STEP SOLUTION

The graph is linear, so its average on [0,4] is the mean of endpoint values: (4+0)/2=2. Integrating gives the same result.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The average is not 3.

✗ **(C)** Underestimates the average.

✗ **(D)** Uses the left endpoint value.

Question 04

★★★ Mandatory

Correct answer: (C) 2/π

STEP-BY-STEP SOLUTION

Average = $(1/\pi)\int_0^{\pi} \sin x\,dx = (1/\pi)[-\cos x]_0^{\pi} = 2/\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the factor 2 from evaluating the antiderivative.

✗ **(B)** This is the integral, not the average.

✗ **(D)** Does not divide by π.

Question 05

★★★ Mandatory

Correct answer: (A) 1/(e−1)

STEP-BY-STEP SOLUTION

$\int_1^e dx/x=1$. Divide by the interval length e−1: average = 1/(e−1).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is the interval length.

✗ **(C)** Uses the upper endpoint reciprocally.

✗ **(D)** Forgets to divide by e−1.

Question 06

★★ Important

Correct answer: (C) 7

STEP-BY-STEP SOLUTION

The average value of a constant function equals that constant.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses the average with a zero change.

✗ **(B)** This is not the constant function value.

✗ **(D)** The interval length is not the average value.

Question 07

★★★ Mandatory

Correct answer: (C) 1

STEP-BY-STEP SOLUTION

$\int_{-1}^{2} x^2 dx = [x^3/3]_{-1}^{2} = 8/3 + 1/3 = 3$. Divide by length 3: average=1.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The integrand is nonnegative and its average is not zero.

✗ **(B)** Uses the integral without dividing by interval length.

✗ **(D)** Uses the upper endpoint value.

Question 08

★★★ Mandatory

Correct answer: (B) 2/(ln 3)

STEP-BY-STEP SOLUTION

The integral is e^(ln3)−e⁰=3−1=2. The interval length is ln3, so the average is 2/ln3.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the initial value only.

✗ **(C)** Multiplies by the interval length instead of dividing.

✗ **(D)** Forgets the factor 2 in the integral.

Question 09

★★★ Mandatory

Correct answer: (A) 0

STEP-BY-STEP SOLUTION

x^3 is odd and the interval is symmetric, so its integral and average are 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** An odd function's symmetric average is zero.

✗ **(C)** Uses the interval endpoint cubed.

✗ **(D)** Uses an endpoint value.

Question 10

★★★★★ Very Important

Correct answer: (C) $4/\sqrt{3}$

STEP-BY-STEP SOLUTION

Average value = $(1/4)\int_0^4 x^2 dx = 16/3$. Solve $c^2=16/3$; in $[0,4]$, $c=4/\sqrt{3}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the average value, not c.

✗ **(B)** Does not solve $c^2=16/3$.

✗ **(D)** This is not the positive square root of 16/3.

Question 11

★★★★★ Very Important

Correct answer: (B) 0

STEP-BY-STEP SOLUTION

$\int_0^4 (t-2)dt=[t^2/2-2t]_0^4=8-8=0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses displacement with total distance.

✗ **(C)** Does not integrate over the full interval.

✗ **(D)** Uses only the positive portion.

Question 12

★★★★★ Very Important

Correct answer: (B) 4

STEP-BY-STEP SOLUTION

The velocity changes sign at t=2. The two triangular areas each have base 2 and height 2, so total distance=2+2=4.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Counts a rectangle instead of the two triangles.

✗ **(C)** Counts only one triangle.

✗ **(D)** Adds an extra area.

Question 13

★★★ Mandatory

Correct answer: (C) −3

STEP-BY-STEP SOLUTION

$\int_0^3(t^2-4)dt=[t^3/3-4t]_0^3=9-12=-3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error.

✗ **(B)** The positive and negative areas do not cancel.

✗ **(D)** This is not the signed integral.

Question 14

★★★★★ Very Important

Correct answer: (D) 23/3

STEP-BY-STEP SOLUTION

$v=0$ at $t=2$. Distance on $[0,2]$ is $\int_0^2(4-t^2)dt=16/3$. On $[2,3]$ it is $\int_2^3(t^2-4)dt=7/3$. Total$=23/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is signed displacement, not total distance.

✗ **(B)** Counts only the negative part.

✗ **(C)** Counts only the positive part.

Question 15

★★★ Mandatory

Correct answer: (C) t=0 and t=2

STEP-BY-STEP SOLUTION

$v=s'=3t^2-6t=3t(t-2)$. Set $v=0$: $t=0$ or $t=2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses $t=2$.

✗ **(B)** Misses $t=0$.

✗ **(D)** These are not the zeros of velocity.

Question 16

★★★ Mandatory

Correct answer: (B) 7

STEP-BY-STEP SOLUTION

$v(2)=v(0)+\int_0^2(6t-4)dt=3+[3t^2-4t]_0^2=3+(12-8)=7.$

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the initial velocity.

✗ **(C)** Adds the full expression without integrating correctly.

✗ **(D)** Arithmetic error in the accumulated change.

Question 17

★★★ Mandatory

Correct answer: (C) −1

STEP-BY-STEP SOLUTION

Use signed trapezoid areas: [0,1] gives 1, [1,2] gives −1, and [2,3] gives −1. Total displacement=−1.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds the three time values.

✗ **(B)** Adds signed areas incorrectly.

✗ **(D)** Treats every area as positive.

Question 18

★★★ Mandatory

Correct answer: (B) −1

STEP-BY-STEP SOLUTION

Average velocity=[s(3)−s(0)]/3=[−3−0]/3=−1.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error.

✗ **(C)** Uses s(3) as the average velocity.

✗ **(D)** Uses displacement without dividing by time.

Question 19

★★★★★ Very Important

Correct answer: (B) 1

STEP-BY-STEP SOLUTION

Factor $v=3(t-1)(t-3)$. It is negative between its roots 1 and 3.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Velocity is not negative for all positive t.

✗ **(C)** The velocity is positive on this interval.

✗ **(D)** Velocity is positive after t=3.

Question 20

★★★★★ Very Important

Correct answer: (C) moving left and slowing down

STEP-BY-STEP SOLUTION

Negative velocity means motion to the left. Opposite signs of velocity and acceleration mean speed is decreasing.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** v is nonzero.

✗ **(B)** Opposite signs mean slowing down.

✗ **(D)** $v<0$ means not moving right.

Question 21

★★★★★ Very Important

Correct answer: (C) x^2+1

STEP-BY-STEP SOLUTION

By the Fundamental Theorem of Calculus, the derivative is the integrand evaluated at x: $F'(x)=x^2+1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is F(x), not F′(x).

✗ **(B)** Not the derivative of the integrand or accumulation function.

✗ **(D)** Differentiates an incorrect expression.

Question 22

★★★★★ Very Important

Correct answer: (D) $2x\cos(x^2)$

STEP-BY-STEP SOLUTION

FTC with chain rule: $G'(x)=\cos(x^2)\cdot 2x$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the chain-rule factor.

✗ **(B)** Wrong sign and wrong function.

✗ **(C)** Uses the derivative of cosine instead of FTC evaluation.

Question 23

★★★★★ Very Important

Correct answer: (B) $-(x^2+1)$

STEP-BY-STEP SOLUTION

The variable is the lower limit, so differentiate with a negative sign: $H'(x)=-(x^2+1)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses the negative sign.

✗ **(C)** Uses the fixed upper limit.

✗ **(D)** Differentiates the integrand.

Question 24

★★★★★ Very Important

Correct answer: (A) $2x\sin(x^2)-\sin(x)$

STEP-BY-STEP SOLUTION

Leibniz rule: $\sin(x^2)\cdot(2x)-\sin(x)\cdot(1)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits the derivative of x^2.

✗ **(C)** The lower-limit term must be subtracted.

✗ **(D)** Differentiates the integrand rather than evaluating it at the limits.

Question 25

★★★ Mandatory

Correct answer: (B) $x^2/2-x$

STEP-BY-STEP SOLUTION

Integrate: $[t^2/2-t]_0^x=x^2/2-x$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Incorrect constant term.

✗ **(C)** Uses an incorrect antiderivative.

✗ **(D)** This is not the accumulated integral.

Question 26

★★★★★ Very Important

Correct answer: (D) $x=-2$ and $x=2$

STEP-BY-STEP SOLUTION

$F'(x)=x^2-4$. Set equal to zero: $x=\pm 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses both zeros of F' except $x=0$, which is not a zero.

✗ **(B)** F' has two zeros.

✗ **(C)** Solves $x^2=16$ instead of 4.

Question 27

★★★★★ Very Important

Correct answer: (D) $x<-2$ or $x>2$

STEP-BY-STEP SOLUTION

$F'(x)=x^2-4$ is positive when $|x|>2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is where F' is negative.

✗ **(B)** F' is negative on $(0,2)$.

✗ **(C)** F' is not positive everywhere.

Question 28

★★ Important

Correct answer: (D) 8

STEP-BY-STEP SOLUTION

Antiderivative x^2; evaluate $9-1=8$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds rather than subtracts endpoint values.

✗ **(B)** Uses only the lower limit.

✗ **(C)** Uses $3\cdot2$.

Question 29

★★ Important

Correct answer: (C) 1

STEP-BY-STEP SOLUTION

Antiderivative $\sin x$; $\sin(\pi/2)-\sin0=1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses interval length with integral.

✗ **(B)** Uses cos at endpoints instead of its antiderivative.

✗ **(D)** Doubles the result.

Question 30

★★★ Mandatory

Correct answer: (D) 32/5

STEP-BY-STEP SOLUTION

Antiderivative $t^5/5$; evaluate $32/5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 2^3.

✗ **(B)** Uses $2^4/5$.

✗ **(C)** Forgets the denominator 5.

Question 31

★★★★★ Very Important

Correct answer: (D) 1/6

STEP-BY-STEP SOLUTION

On [0,1], x is above x^2. Area=$\int_0^1(x-x^2)dx=1/2-1/3=1/6$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only $\int x$.

✗ **(B)** Forgets part of the subtraction.

✗ **(C)** Uses only $\int x^2$ incorrectly.

Question 32

★★★ Mandatory

Correct answer: (D) 4/3

STEP-BY-STEP SOLUTION

$2x$ is above x^2 on [0,2]. Area=$\int_0^2(2x-x^2)dx=4-8/3=4/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the first integral.

✗ **(B)** Subtracts the terms incorrectly.

✗ **(C)** Uses endpoint heights as area.

Question 33

★★★ Mandatory

Correct answer: (A) 32/3

STEP-BY-STEP SOLUTION

Area=$\int_{-2}^{2}(4-x^2)dx=16-16/3=32/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses only half the interval.

✗ **(C)** Uses rectangle area 4×4 without subtracting the parabola.

✗ **(D)** Forgets the constant area contribution.

Question 34

★★★★★ Very Important

Correct answer: (D) 9/2

STEP-BY-STEP SOLUTION

The line is above the parabola. Area=$\int_{-1}^{2}(-x^2+x+2)dx=[-x^3/3+x^2/2+2x]_{-1}^{2}=9/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error.

✗ **(B)** Uses a bounding rectangle instead.

✗ **(C)** Underestimates the integral.

Question 35

★★★ Mandatory

Correct answer: (A) 4/3

STEP-BY-STEP SOLUTION

$\sqrt{x} \geq x/2$ on $[0,4]$. Area=$\int_0^4(\sqrt{x}-x/2)dx=16/3-4=4/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Doubles an incorrect result.

✗ **(C)** Uses only $\int\sqrt{x}$.

✗ **(D)** Misses the x/2 contribution.

Question 36

★★★ Mandatory

Correct answer: (C) 1/4

STEP-BY-STEP SOLUTION

$x \geq x^3$ on $[0,1]$. Area=$\int_0^1(x-x^3)dx=1/2-1/4=1/4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only $\int x$.

✗ **(B)** Incorrect subtraction.

✗ **(D)** Uses only $\int x^3$.

Question 37

★★★ Mandatory

Correct answer: (B) 4/3

STEP-BY-STEP SOLUTION

Area=$\int_{-1}^{1}(1-x^2)dx=2-2/3=4/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the interval is symmetric and misses the correct constant integral.

✗ **(C)** Underestimates the result.

✗ **(D)** Uses the rectangle area only.

Question 38

★★★★★ Very Important

Correct answer: (A) 9/2

STEP-BY-STEP SOLUTION

The line is above the parabola on $[-2,1]$. Area=$\int_{-2}^{1}(2-x-x^2)dx=9/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds areas with an incorrect sign.

✗ **(C)** Overestimates the result.

✗ **(D)** Arithmetic error in the antiderivative.

Question 39

★★★ Mandatory

Correct answer: (B) 9/2

STEP-BY-STEP SOLUTION

$3x\geq x^2$ on $[0,3]$. Area=$\int_{0}^{3}(3x-x^2)dx=27/2-9=9/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the linear triangle area.

✗ **(C)** Subtracts incorrectly.

✗ **(D)** Misses half of one term.

Question 40

★★★ Mandatory

Correct answer: (A) 1

STEP-BY-STEP SOLUTION

The region is two right triangles, each area 1/2; total area=1.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds an extra half-unit.

✗ **(C)** Counts only one of the two triangles.

✗ **(D)** Uses the rectangle area.

Question 41

★★ Important

Correct answer: (B) 1

STEP-BY-STEP SOLUTION

$\int_0^{(\pi/2)} \cos x\,dx = \sin(\pi/2) - \sin 0 = 1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The function is positive on the interval.

✗ **(C)** Confuses area with interval length.

✗ **(D)** Doubles the integral.

Question 42

★★ Important

Correct answer: (B) 2

STEP-BY-STEP SOLUTION

$\text{Area} = \int_0^2 (2-x)\,dx = 4 - 2 = 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the rectangle area without subtracting.

✗ **(C)** Uses only the rectangle area.

✗ **(D)** Underestimates the triangle.

Question 43

★★ Important

Correct answer: (C) 8

STEP-BY-STEP SOLUTION

Area=$\int_0^4(4-x)dx=16-8=8$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the full rectangle and ignores the triangular gap.

✗ **(B)** Underestimates the exact area.

✗ **(D)** Underestimates the triangular region.

Question 44

★★★ Mandatory

Correct answer: (A) 9

STEP-BY-STEP SOLUTION

The curves meet at x=3. Area=$\int_0^3[(6-x)-x]dx=\int_0^3(6-2x)dx=18-9=9$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Misses half of the area.

✗ **(C)** Uses a rectangle with an incorrect height.

✗ **(D)** Uses the integral of 6−x only.

Question 45

★★★ Mandatory

Correct answer: (B) $2\sqrt{3}$

STEP-BY-STEP SOLUTION

Area=$\int_0^\sqrt{3}(3-x^2)dx=3\sqrt{3}-(3\sqrt{3})/3=2\sqrt{3}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses the subtraction term.

✗ **(C)** Uses only the rectangle area.

✗ **(D)** Doubles the correct result.

Question 46

★★★★★ Very Important

Correct answer: (B) 4/3

STEP-BY-STEP SOLUTION

Right minus left is $2y-y^2$. Area$=\int_0^2(2y-y^2)dy=4-8/3=4/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error.

✗ **(C)** Uses only the second integral.

✗ **(D)** Uses only the first integral.

Question 47

★★★ Mandatory

Correct answer: (B) 1/6

STEP-BY-STEP SOLUTION

Right minus left is $y-y^2$. Integral$=1/2-1/3=1/6$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only $\int y$.

✗ **(C)** Uses only $\int y^2$.

✗ **(D)** Forgets one term.

Question 48

★★★ Mandatory

Correct answer: (B) 32/3

STEP-BY-STEP SOLUTION

Area$=\int_{-2}^{2}(4-y^2)dy=16-16/3=32/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses part of the interval.

✗ **(C)** Uses half the interval.

✗ **(D)** Uses rectangle area only.

Question 49

★★★ Mandatory

Correct answer: (D) $2\sqrt{3}$

STEP-BY-STEP SOLUTION

Right minus left is $3-y^2$. Area=$3\sqrt{3}-(\sqrt{3})^3/3=3\sqrt{3}-\sqrt{3}=2\sqrt{3}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Doubles the correct result.

✗ **(B)** Uses only the rectangle area.

✗ **(C)** Misses one contribution.

Question 50

★★ Important

Correct answer: (A) 2

STEP-BY-STEP SOLUTION

Area=$\int_0^2(2-y)dy=4-2=2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses an incorrect antiderivative.

✗ **(C)** Uses the rectangle area.

✗ **(D)** Underestimates the triangle.

Question 51

★★ Important

Correct answer: (A) 8

STEP-BY-STEP SOLUTION

Area=$\int_0^4(4-y)dy=16-8=8$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Underestimates the region.

✗ **(C)** Underestimates the result.

✗ **(D)** Uses the full 4-by-4 rectangle.

Question 52

★★★ Mandatory

Correct answer: (A) 16/3

STEP-BY-STEP SOLUTION

Area=$\int_0^4 \sqrt{y}\, dy$=(2/3)·4^(3/2)=16/3.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses rectangle area.

✗ **(C)** Uses y instead of √y.

✗ **(D)** Incorrect antiderivative.

Question 53

★★ Important

Correct answer: (B) 4

STEP-BY-STEP SOLUTION

Area=$\int_0^2 (1+y)dy$=2+2=4.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Underestimates the area.

✗ **(C)** Adds endpoints instead of integrating.

✗ **(D)** Misses the y contribution.

Question 54

★★★ Mandatory

Correct answer: (D) 6√3

STEP-BY-STEP SOLUTION

The horizontal width is 3 and the vertical interval height is 2√3, so area=3·2√3=6√3.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Underestimates the area.

✗ **(B)** This is not the full width times height.

✗ **(C)** Overestimates the area.

Question 55

★★★★★ Very Important

Correct answer: (A) 9/2

STEP-BY-STEP SOLUTION

On this interval $y+2 \geq y^2$. Area $= \int_{-1}^{2}(y+2-y^2)dy = [y^2/2+2y-y^3/3]_{-1}^{2} = 9/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Overestimates the integral.

✗ **(C)** Uses an incorrect order of subtraction.

✗ **(D)** Arithmetic error.

Question 56

★★★★★ Very Important

Correct answer: (B) 8

STEP-BY-STEP SOLUTION

Cross-sectional area $A(x)=(\sqrt{x})^2=x$. Volume $= \int_0^4 x\,dx = 8$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses side length squared incorrectly as x^2.

✗ **(C)** Uses the interval length only.

✗ **(D)** Arithmetic error.

Question 57

★★★ Mandatory

Correct answer: (D) 8/3

STEP-BY-STEP SOLUTION

$A(x)=(2-x)^2$. Volume $= \int_0^2 (2-x)^2 dx = 8/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses half the correct integral.

✗ **(B)** Uses a rectangular prism bound.

✗ **(C)** Uses an incorrect square-area expression.

Question 58

★★★★★ Very Important

Correct answer: (D) 16/15

STEP-BY-STEP SOLUTION

$V=\int_{-1}^{1}(1-x^2)^2dx=2\int_0^1(1-2x^2+x^4)dx=2(1-2/3+1/5)=16/15.$

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses a factor of 2.

✗ **(B)** Uses an upper bound instead of integrating.

✗ **(C)** Uses the wrong cross-sectional area.

Question 59

★★★ Mandatory

Correct answer: (C) 2/3

STEP-BY-STEP SOLUTION

Cross-sectional area = $x(2x)=2x^2$. $V=\int_0^1 2x^2dx=2/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the endpoint area as volume.

✗ **(B)** Forgets the factor 2 in the height.

✗ **(D)** Uses $\int 3x^2$.

Question 60

★★★ Mandatory

Correct answer: (C) √3/12

STEP-BY-STEP SOLUTION

Area of an equilateral triangle is $(\sqrt{3}/4)s^2$. $V=(\sqrt{3}/4)\int_0^1(1-x)^2dx=(\sqrt{3}/4)(1/3)=\sqrt{3}/12$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the triangle area coefficient.

✗ **(B)** Misses a factor of 2.

✗ **(D)** Doubles the exact volume.

Question 61

★★★★★ Very Important

Correct answer: (C) π/3

STEP-BY-STEP SOLUTION

A semicircle with diameter d has area $\pi d^2/8$. $V=(\pi/8)\int_0^2(2-x)^2dx=(\pi/8)(8/3)=\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses a factor of 2.

✗ **(B)** Uses a full-circle area.

✗ **(D)** Uses the incorrect radius/diameter conversion.

Question 62

★★★★★ Very Important

Correct answer: (A) π/6

STEP-BY-STEP SOLUTION

Area$=(\pi/8)d^2=(\pi/8)(1-x^2)$. Integrate: $(\pi/8)(2-2/3)=\pi/6$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Doubles the correct volume.

✗ **(C)** Misses the factor from the integral.

✗ **(D)** Uses full-circle cross sections.

Question 63

★★★ Mandatory

Correct answer: (C) 4/3

STEP-BY-STEP SOLUTION

For a right-isosceles triangle with leg b, area$=b^2/2$. $V=(1/2)\int_0^2(2-x)^2dx=(1/2)(8/3)=4/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses a factor of 2.

✗ **(B)** Uses square cross sections.

✗ **(D)** Uses only the base integral.

Question 64

★★★ Mandatory

Correct answer: (C) √3/3

STEP-BY-STEP SOLUTION

$A(x)=(\sqrt{3}/4)(2x)^2=\sqrt{3}x^2$. Integrating gives √3/3.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses area at x=1 as the volume.

✗ **(B)** Doubles the integral.

✗ **(D)** Uses side x instead of 2x.

Question 65

★★★ Mandatory

Correct answer: (A) 1

STEP-BY-STEP SOLUTION

$A(x)=3(1-x)^2$. $V=3\int_0^1(1-x)^2dx=3(1/3)=1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Misses a factor of 2.

✗ **(C)** Uses endpoint height instead of integrating.

✗ **(D)** Uses $3\int(1-x)$.

Question 66

★★★ Mandatory

Correct answer: (B) 4/3

STEP-BY-STEP SOLUTION

$A(x)=(2x)^2=4x^2$. $V=\int_0^1 4x^2dx=4/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses endpoint area only.

✗ **(C)** Uses $2x^2$.

✗ **(D)** Uses x^2 instead of $4x^2$.

Question 67

★★★ Mandatory

Correct answer: (C) π/4

STEP-BY-STEP SOLUTION

$A=(\pi/8)d^2=(\pi/8)(4x)=\pi x/2$. Integrating 0 to1 gives π/4.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses a factor of 2.

✗ **(B)** Doubles the result.

✗ **(D)** Uses full-circle cross sections.

Question 68

★★★★★ Very Important

Correct answer: (D) 32/3

STEP-BY-STEP SOLUTION

Area$=(\sqrt{(4-x^2)})^2=4-x^2$. Volume$=\int_{-2}^{2}(4-x^2)dx=[4x-x^3/3]_{-2}^{2}=16-16/3=32/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses bounding rectangle 4×4 without subtracting.

✗ **(B)** Counts only one half of the region.

✗ **(C)** Integrates over [0,2] only.

Question 69

★★★★★ Very Important

Correct answer: (A) 8π/3

STEP-BY-STEP SOLUTION

Disk method: $V=\pi\int_0^2 x^2dx=\pi(8/3)=8\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses an incorrect radius integral.

✗ **(C)** Doubles the exact result.

✗ **(D)** Uses endpoint radius squared times interval length.

Question 70

★★★★★ Very Important

Correct answer: (B) 8π

STEP-BY-STEP SOLUTION

$V=\pi\int_0^4(\sqrt{x})^2dx=\pi\int_0^4 x\,dx=8\pi.$

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $\int\sqrt{x}$ rather than $\int x$.

✗ **(C)** Uses a rectangular prism bound.

✗ **(D)** Uses half the integral.

Question 71

★★★ Mandatory

Correct answer: (C) 18π

STEP-BY-STEP SOLUTION

$V=\pi r^2L=\pi(3^2)(2)=18\pi.$

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses area coefficient incorrectly.

✗ **(B)** Uses radius 3 times length.

✗ **(D)** Doubles the volume.

Question 72

★★★★★ Very Important

Correct answer: (D) 16π/15

STEP-BY-STEP SOLUTION

$V=\pi\int_{-1}^{1}(1-x^2)^2dx=\pi(16/15)=16\pi/15.$

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses a factor of 2.

✗ **(B)** Uses a different power expansion.

✗ **(C)** Uses an incorrect rectangular approximation.

Question 73

★★★★★ Very Important

Correct answer: (D) 8π

STEP-BY-STEP SOLUTION

Horizontal disks have radius $\sqrt{y}$. $V=\pi\int_0^4 y\,dy=8\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $\int\sqrt{y}$.

✗ **(B)** Uses radius squared incorrectly.

✗ **(C)** Uses half the correct integral.

Question 74

★★★★★ Very Important

Correct answer: (A) 32π/5

STEP-BY-STEP SOLUTION

Disk radius is y^2. $V=\pi\int_0^2 (y^2)^2 dy=\pi\int_0^2 y^4 dy=32\pi/5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses an incorrect antiderivative.

✗ **(C)** Uses $\int y^2$.

✗ **(D)** Forgets to square the radius correctly.

Question 75

★★ Important

Correct answer: (D) π/3

STEP-BY-STEP SOLUTION

$V=\pi\int_0^1 x^2 dx=\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses half of $\int x^2$.

✗ **(B)** Uses endpoint area only.

✗ **(C)** Uses $\int x$.

Question 76

★★★ Mandatory

Correct answer: (C) 4π/3

STEP-BY-STEP SOLUTION

$V=\pi\int_0^1(2x)^2dx=4\pi\int_0^1x^2dx=4\pi/3.$

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses an incorrect factor.

✗ **(B)** Uses radius x rather than 2x.

✗ **(D)** Uses endpoint radius alone.

Question 77

★★ Important

Correct answer: (B) 48π

STEP-BY-STEP SOLUTION

$V=\pi(4^2)(3)=48\pi.$

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses half the length.

✗ **(C)** Uses radius instead of radius squared.

✗ **(D)** Uses radius 4 and length 4.

Question 78

★★★★★ Very Important

Correct answer: (C) 14π/3

STEP-BY-STEP SOLUTION

Disk radius=1+y/2. $V=\pi\int_0^2(1+y/2)^2dy=\pi[\ y+y^2/2+y^3/12\ ]_0^2=14\pi/3.$

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses an incorrect antiderivative.

✗ **(B)** Overestimates the integral.

✗ **(D)** Arithmetic error in the cubic term.

Question 79

★★★★★ Very Important

Correct answer: (A) $32\sqrt{2}\pi/15$

STEP-BY-STEP SOLUTION

$V=\pi\int_0^{\sqrt{2}}(2-x^2)^2dx=\pi[4x-4x^3/3+x^5/5]_0^{\sqrt{2}}=32\sqrt{2}\pi/15$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses a rough rectangle approximation.

✗ **(C)** Uses the endpoint radius only.

✗ **(D)** Misses a factor of 2.

Question 80

★★★★★ Very Important

Correct answer: (D) $16\pi/3$

STEP-BY-STEP SOLUTION

Washer radii $R=2$ and $r=x$. $V=\pi\int_0^2(4-x^2)dx=\pi(8-8/3)=16\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to subtract the inner disk.

✗ **(B)** Uses only the inner-radius contribution incorrectly.

✗ **(C)** Uses only outer radius times interval length.

Question 81

★★★★★ Very Important

Correct answer: (C) $\pi/6$

STEP-BY-STEP SOLUTION

Outer radius $\sqrt{x}$, inner radius x. $V=\pi\int_0^1(x-x^2)dx=\pi(1/2-1/3)=\pi/6$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $\int\sqrt{x}$ rather than the washer formula.

✗ **(B)** Uses $\int x$ only.

✗ **(D)** Misses the difference of radii squared.

Question 82

★★★ Mandatory

Correct answer: (D) 16π

STEP-BY-STEP SOLUTION

$V=\pi\int_0^2(3^2-1^2)dx=\pi(8)(2)=16\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the outer disk.

✗ **(B)** Adds the squared radii.

✗ **(C)** Uses radius difference rather than difference of squares.

Question 83

★★★★★ Very Important

Correct answer: (D) 64π/15

STEP-BY-STEP SOLUTION

Outer radius 2x, inner radius x^2. $V=\pi\int_0^2(4x^2-x^4)dx=\pi(32/3-32/5)=64\pi/15$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses a factor of 2.

✗ **(B)** Forgets the inner-radius fourth-power term.

✗ **(C)** Uses disk method without subtracting the inner radius.

Question 84

★★★ Mandatory

Correct answer: (B) 56π

STEP-BY-STEP SOLUTION

$V=\pi\int_0^4(16-x)dx=\pi(64-8)=56\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds the inner contribution.

✗ **(C)** Uses a wrong inner-radius integral.

✗ **(D)** Does not subtract the inner volume.

Question 85

★★★★★ Very Important

Correct answer: (C) 4π/3

STEP-BY-STEP SOLUTION

Outer radius is x+1 and inner radius is 1. $V=\pi\int_0^1[(x+1)^2-1]dx=\pi\int_0^1(x^2+2x)dx=4\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses outer disks but forgets the hole.

✗ **(B)** Uses only part of the washer area.

✗ **(D)** Misses a factor of 2.

Question 86

★★★★★ Very Important

Correct answer: (A) 28π/3

STEP-BY-STEP SOLUTION

Outer radius=3, inner radius=3−x. $V=\pi\int_0^2[9-(3-x)^2]dx=\pi\int_0^2(6x-x^2)dx=28\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses only the full outer cylinders.

✗ **(C)** Arithmetic error.

✗ **(D)** Underestimates the integral.

Question 87

★★★★★ Very Important

Correct answer: (B) 16π/3

STEP-BY-STEP SOLUTION

Washer radii R=2 and r=y. $V=\pi\int_0^2(4-y^2)dy=16\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses disk method with radius 2 and ignores the hole.

✗ **(C)** Doubles an incorrect result.

✗ **(D)** Uses only the inner radius.

Question 88

★★★★★ Very Important

Correct answer: (A) 5π/3

STEP-BY-STEP SOLUTION

Outer radius 2, inner radius 1+y. $V=\pi\int_0^1[4-(1+y)^2]dy=\pi\int_0^1(3-2y-y^2)dy=5\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses outer disks only.

✗ **(C)** Doubles the correct result.

✗ **(D)** Arithmetic error in integrating y^2.

Question 89

★★★★★ Very Important

Correct answer: (D) 8π/5

STEP-BY-STEP SOLUTION

Washer radii R=1 and $r=x^2$. $V=\pi\int_{-1}^1(1-x^4)dx=\pi(2-2/5)=8\pi/5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the inner hole.

✗ **(B)** Misses a factor of 2.

✗ **(C)** Uses only the unit-radius disk volume over length 1.

Question 90

★★★ Mandatory

Correct answer: (C) 9π

STEP-BY-STEP SOLUTION

$V=\pi\int_0^3(2^2-1^2)dx=\pi(3)(3)=9\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only part of the interval.

✗ **(B)** Adds the squared radii instead of subtracting.

✗ **(D)** Uses radius difference rather than difference of squares.

Question 91: Area and Net Area from a Graph

★★★★★ Very Important

MODEL SOLUTION

(a) Vertices are (0,0), (2,4), (5,−2), (8,0). From 0 to 2 the triangle area is +4. The line from (2,4) to (5,−2) crosses at x=4: area from 2 to 4 is +4 and from 4 to 5 is −1. From 5 to 8 the triangle is below the axis, area −3. Integral =4+4−1−3=**4**.
(b) Total area=4+4+1+3=**12**.
(c) Average value=4/8=**1/2**.
(d) The definite integral is signed; below-axis areas subtract. Total geometric area counts all regions positively.

SCORING RUBRIC

Criterion	Points
Correct signed areas for graph regions	2
Net integral equals 4	1
Total area equals 12	2
Average value equals 1/2	2
Explains signed vs. total area	2
Total	9

Question 92: Average Value and the Integral Mean Value Theorem

★★★★★ Very Important

MODEL SOLUTION

(a) Average value=$(1/3)\int_0^3 x^2 dx=(1/3)(9)=\mathbf{3}$.

(b) Solve $c^2=3$. Since $c\in[0,3]$, $\mathbf{c=\sqrt{3}}$.

(c) f is continuous on the closed interval [0,3]. The Integral Mean Value Theorem guarantees a c in [0,3] where f(c) equals the average value.

SCORING RUBRIC

Criterion	Points
Correct average-value formula and integral	2
Average value 3	1
Solves $c^2=3$	2
Selects $c=\sqrt{3}$ in interval	1
States continuity and theorem conclusion	3
Total	9

Question 93: Motion: Displacement and Total Distance

★★★★★ Very Important

MODEL SOLUTION

(a) $\int_0^5 (t^2-4t+3)dt = [t^3/3 - 2t^2 + 3t]_0^5 = 125/3 - 50 + 15 = \mathbf{20/3\ m}$.

(b) $v=(t-1)(t-3)$, so the sign changes at 1 and 3. Signed pieces: $\int_0^1 v = 4/3$, $\int_1^3 v = -4/3$, $\int_3^5 v = 20/3$. Total distance $= 4/3 + 4/3 + 20/3 = \mathbf{28/3\ m}$.

(c) $v>0$ on **$[0,1)$ and $(3,5]$**.

SCORING RUBRIC

Criterion	Points
Correct antiderivative and displacement	2
Displacement 20/3 m	1
Finds velocity zeros 1 and 3	1
Splits the total-distance integral correctly	2
Distance 28/3 m	1
Positive-direction intervals	1
Total	9

Question 94: Area Between Curves

★★★★★ Very Important

MODEL SOLUTION

(a) $2x-x^2=x(2-x)>0$ for $0<x<2$, so $y=2x$ is above $y=x^2$.

(b) Area$=\int_0^2(2x-x^2)dx=[x^2-x^3/3]_0^2=4-8/3=$**4/3** square units.

(c) Average vertical distance=area/$(2-0)=(4/3)/2=$**2/3**.

SCORING RUBRIC

Criterion	Points
Correct comparison with justification	2
Correct area integral setup	2
Correct area 4/3	2
Average vertical distance setup	1
Correct average 2/3	2
Total	9

Question 95: Volume with Square Cross Sections

★★★★★ Very Important

MODEL SOLUTION

(a) The vertical base segment runs from $y=0$ to $y=\sqrt{x}$, so its length is $\sqrt{x}$.

(b) A square with side $\sqrt{x}$ has area $A(x)=(\sqrt{x})^2=\mathbf{x}$.

(c) $V=\int_0^4 A(x)dx=\int_0^4 x\,dx=[x^2/2]_0^4=$ **8 cubic units**.

SCORING RUBRIC

Criterion	Points
Correct side length $\sqrt{x}$	2
Correct area $A(x)=x$	2
Correct definite-integral setup	2
Correct antiderivative/evaluation	2
Correct volume 8	1
Total	9

Question 96: Volume with Semicircular Cross Sections

★★★★★ Very Important

MODEL SOLUTION

(a) Diameter $d=\sqrt{(1-x^2)}$; radius $r=\sqrt{(1-x^2)}/2$.

(b) Semicircle area $=(1/2)\pi r^2=\pi(1-x^2)/8$.

(c) $V=\int_{-1}^{1}\pi(1-x^2)/8\,dx=(\pi/8)(2-2/3)=$ **$\pi/6$ cubic units**.

SCORING RUBRIC

Criterion	Points
Correct diameter and radius	2
Correct semicircle-area formula	2
Correct $A(x)=\pi(1-x^2)/8$	1
Correct volume integral	2
Correct result $\pi/6$	2
Total	9

Question 97: Disk Method About the x-Axis

★★★★★ Very Important

MODEL SOLUTION

(a) The disk radius is $R(x)=\sqrt{x}$.

(b) $V=\pi\int_0^4 R(x)^2\,dx=\pi\int_0^4 x\,dx$.

(c) $V=\pi[x^2/2]_0^4=$ **8π cubic units**.

SCORING RUBRIC

Criterion	Points
Correct radius $\sqrt{x}$	2
Correct disk-method setup	3
Correct antiderivative	2
Correct result 8π	2
Total	9

Question 98: Washer Method About the x-Axis

★★★★★ Very Important

MODEL SOLUTION

(a) Outer radius $R=2$; inner radius $r=x$.

(b) $V=\pi\int_0^2(R^2-r^2)dx=\pi\int_0^2(4-x^2)dx$.

(c) $V=\pi[4x-x^3/3]_0^2=\pi(8-8/3)=$**$16\pi/3$ cubic units**.

SCORING RUBRIC

Criterion	Points
Correct outer radius 2 and inner radius x	2
Correct washer formula/setup	3
Correct antiderivative and limits	2
Correct result $16\pi/3$	2
Total	9

Question 99: Volume with Cross Sections from a Table

★★★ Mandatory

MODEL SOLUTION

(a) $\Delta x=1$. Square area is s^2. Left estimate$=1[(1)^2+(2)^2+(1)^2]=$**6 cubic units**.
(b) Right estimate$=1[(2)^2+(1)^2+(0)^2]=$**5 cubic units**.
(c) The volume integrand is the area of each square cross section. A square of side s has area $s\cdot s=s^2$; s alone is only a length.

SCORING RUBRIC

Criterion	Points
Uses square cross-sectional area s^2	1
Correct left-sum setup and value 6	2
Correct right-sum setup and value 5	3
Explains length versus area	3
Total	9

Question 100: Washer Method About a Shifted Axis

★★★★★ Very Important

MODEL SOLUTION

(a) The axis $y=3$ is above the region. Outer radius is from $y=3$ to $y=0$: $R=3$. Inner radius is from $y=3$ to $y=x$: $r=3-x$.

(b) $V=\pi\int_0^2[R^2-r^2]dx=\pi\int_0^2[9-(3-x)^2]dx$.

(c) The integrand simplifies to $6x-x^2$. $V=\pi[3x^2-x^3/3]_0^2=\pi(12-8/3)=$**$28\pi/3$ cubic units**.

SCORING RUBRIC

Criterion	Points
Correct outer and inner radii	2
Correct washer setup	3
Correct simplification and antiderivative	2
Correct result $28\pi/3$	2
Total	9

Essential Formulas & Integration Theorems

- **Average value:** $f_{avg} = (1/(b-a))\int_a^b f(x)dx$.
- **Net change theorem:** accumulated change = $\int_a^b$ rate(t)dt.
- **Motion:** displacement = $\int v(t)dt$; total distance = $\int |v(t)|dt$; speed = $|v(t)|$.
- **Area between curves:** $\int$ (top − bottom)dx, or $\int$ (right − left)dy.
- **Known cross sections:** $V = \int A(x)dx$. Squares: s^2; Rectangles: $b \cdot h$; Semicircles: $(\pi/8)d^2$; Equilateral triangles: $(\sqrt{3}/4)s^2$; Isosceles right triangles: $(1/2)leg^2$.
- **Disk method:** $V = \pi\int R(x)^2\,dx$ (about horizontal axis) or $\pi\int R(y)^2\,dy$ (about vertical axis).
- **Washer method:** $V = \pi\int [R^2 - r^2]\,dx$ or $\pi\int [R^2 - r^2]\,dy$, where R = outer radius and r = inner radius.
- **Shifted axis of revolution:** radius is always distance from axis to boundary curve: $|y_{axis} - y_{curve}|$ or $|x_{axis} - x_{curve}|$.

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